

Program : Diploma in Artificial Intelligence & Machine Learning	
Course Code : 4343	Course Title: Object Oriented Programming Concepts using Java
Semester : 4	Credits: 4
Course Category: Program Core	
Periods per week: 4 (L:4 T:0 P:0)	Periods per semester: 60

Course Objectives:

- Students should be able to understand the concepts of object oriented programming and develop simple JAVA applications. .

Course Prerequisites:

Topic	Course name	Semester
Knowledge in Procedural programming language	Problem Solving and Programming	2

Course Outcomes

On completion of the course, the students will be able to

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Describe Object oriented programming paradigms and familiarize Java environments.	12	Understanding
CO2	Develop simple Java Programs.	14	Applying
CO3	Implement inheritance and polymorphism in Java	16	Applying
CO4	Demonstrate Packages, Exception Handling and GUI programming in Java.	16	Applying
	Series Test	2	

CO-PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3						
CO3	3						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Describe Object oriented programming paradigms and familiarize Java environments		
M1.01	Explain the features of Object-Oriented programming	4	Understanding
M1.02	Familiarize the features of java	2	Understanding
M1.03	Understand Java Programming Environment and Runtime Environment	2	Understanding
M1.04	Understand development Platforms -Standard, Enterprise, Java Virtual Machine (JVM), Java compiler and Byte Code.	2	Understanding
M1.05	Outline the general structure of a Java program and explain steps to build Java application	2	Understanding
Contents: Object-Oriented: Characteristics- classes, objects, methods, encapsulation, abstraction, inheritance, dynamic binding, polymorphism. Overview of Java: Features of Java, Java programming Environment and Runtime Environment, Development Platforms -Standard, Enterprise, Java Virtual Machine (JVM), Java compiler, Byte code. Java Program Structure, Steps to build java applications.			
CO2	Develop simple Java Programs.		
M2.01	Identify the data types, variables, Type Conversion and Casting, Arrays in Java.	2	Understanding

M2.02	Describe the Arithmetic Operators, Bitwise Operators, Relational Operators, Boolean Logical Operators, Assignment Operator, Conditional (Ternary) Operator, and Operator Precedence in java	2	Understanding
M2.03	Demonstrate the creation & implementation of classes and objects in Java.	3	Applying
M2.04	Explain Visibility Modifiers	1	Understanding
M2.05	Create Objects using constructors.	2	Applying
M2.06	Illustrate reading data from console and files, Writing data to console and files in Java	2	Applying
M2.07	Implement the Control Structures in Java	2	Applying
	Series Test – I	1	

Contents:

Core Java Fundamentals: Data types, variables, Type Conversion and Casting, Arrays. Operators - Arithmetic Operators, Bitwise Operators, Relational Operators, Boolean Logical Operators, Assignment Operator, Conditional (Ternary) Operator, and Operator Precedence. Classes, objects, methods, visibility modifiers, constructors, constructor overloading Input / Output operations -Reading and Writing data to console and files. Control Structures - Selection Statements, Iteration Statements and Jump Statements.

CO3	Implement inheritance and polymorphism in Java		
M3.01	Describe inheritance and types	1	Understanding
M3.02	Illustrate the invoking of superclass constructors	2	Applying
M3.03	Illustrate the invoking of superclass methods	2	Applying
M3.04	Explain polymorphism , dynamic binding and visibility controls	2	Understanding
M3.05	Illustrate overriding	3	Applying
M3.06	Demonstrate final variables, final methods, final classes, abstract methods and abstract classes.	3	Applying
M3.07	Demonstrate interfaces	3	Applying

Contents:

Inheritance – Types of Inheritance – Superclass and subclass – calling super class constructor and methods – Polymorphism – dynamic binding – Visibility controls. Overloading vs Overriding - Final variables, final methods and final classes – Abstract methods and class. Interfaces-Definition- Extending Interfaces, Implementing Interfaces.

CO4	Demonstrate Packages, Exception Handling and GUI programming		
M4.01	Develop Java API Packages, System packages, Naming conventions, Creation and usage of packages.	3	Applying
M4.02	Demonstrate Exception Handling in JAVA	3	Applying
M4.03	Implement GUI programming with java applets.	5	Applying
M4.04	Illustrate event handling in java applets.	5	Applying
	Series Test – II	1	

Contents:

Packages - Java API Packages, System packages, Naming conventions, Creating packages, Accessing Packages, Using a Package

Exceptions -Types of Exceptions - Exception handling-try catch, throw,throws

Java Applets-java applets basics, life cycle, creating GUI applications with applets and event handling.

Text / Reference:

T/R	Book Title / Author
T1	Balaguruswamy E: Programming with Java, 6th edition.
T2	Herbert Schildt, “Java the complete reference”, McGraw Hill, Osborne, 7th Edition, 2011
R1	Herbert Schildt and Dale Skrien, ”Java Fundamentals – A comprehensive Introduction”, McGraw Hill, 1st Edition, 2013.
R2	T.Budd, “Understanding Object- Oriented Programming with Java”, Pearson Education, Updated Edition (New Java 2 Coverage), 1999

Online Resources:

Sl. No.	Website Link
1	http://java.sun.com 2. http://www.oracle.com/technetwork/java/index.html)
2	https://www.tutorialspoint.com/java/index.htm
3	http://download.oracle.com/javase/7/docs/api/index.html
4	http://docs.oracle.com/javase/tutorial/iiti.ac.in/people/~tanimad/JavaTheCompleteReference.pdf